

OUT-OF-DISTRIBUTION AND ANOMALY DETECTION

IA Frameworks

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LEARNING OUTCOMES

- Master the fundamental OOD / AD concepts
- Understand the main algorithm families, their advantages and weaknesses
- Familiarize with the most common evaluation metrics

WHAT IS AN ANOMALY?

Anomaly

"An observation that deviates considerably from some concept of normality"

WHY CARE ABOUT ANOMALY DETECTION?

- Intrusion detection (cybersecurity)
- Fraud detection (finance)
- Industrial fault / damage detection
- Medical diagnosis
- **[NEW!]** Out-of-Distribution detection for ML and DL models

MAIN CHALLENGES

- Ill-defined problem
- Absence of abnormal data
- Deep Learning use-cases:
data is very high-dimensional

WHAT IS AN ANOMALY?

"An observation that deviates considerably from some concept of normality"

A typical mathematical formalization:

$\mathcal{X} \subset \mathbb{R}^D$ data space

$\mathbb{P}^+$ ground-truth law of normal data on $\mathcal{X}$ with p.d.f. p^+

$A_z = \{x \in \mathcal{X} \mid p^+(x) \leq z\} \leftarrow \text{set of anomalies}$


CAREFUL!

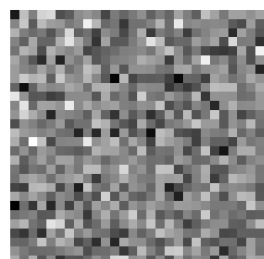
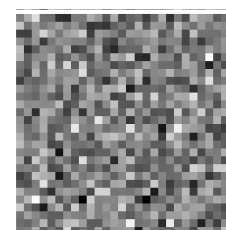
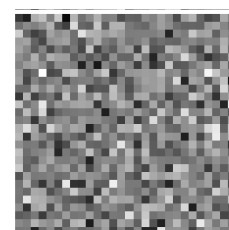
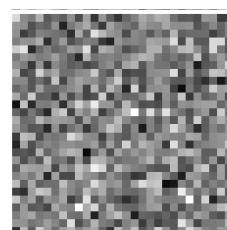
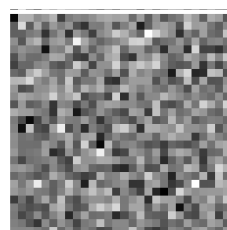
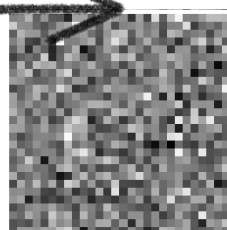
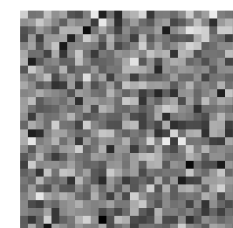
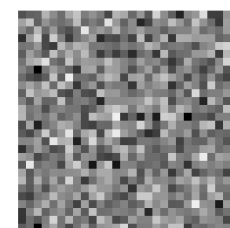
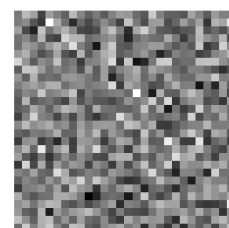
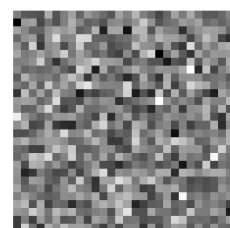
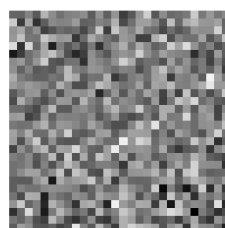
What is the concept of normality used in this formalization ?

CAREFUL!

What is the concept of normality used in this formalization?

10 NORMAL

images



Question: which of these 2 images is normal?

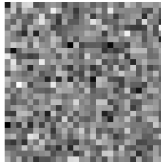
CAREFUL!

Yet: pixels are i.i.d.

$\mathcal{N}(0,1)$ random variables :

$\Rightarrow$ Likelihood Ratio

$$\frac{f(\text{smooth})}{f(\text{noisy})} = \prod_{ij} \frac{1}{e^{-\frac{x_{ij}^2}{2}}}$$

value of pixel ij
in 

TYPES OF ANOMALIES

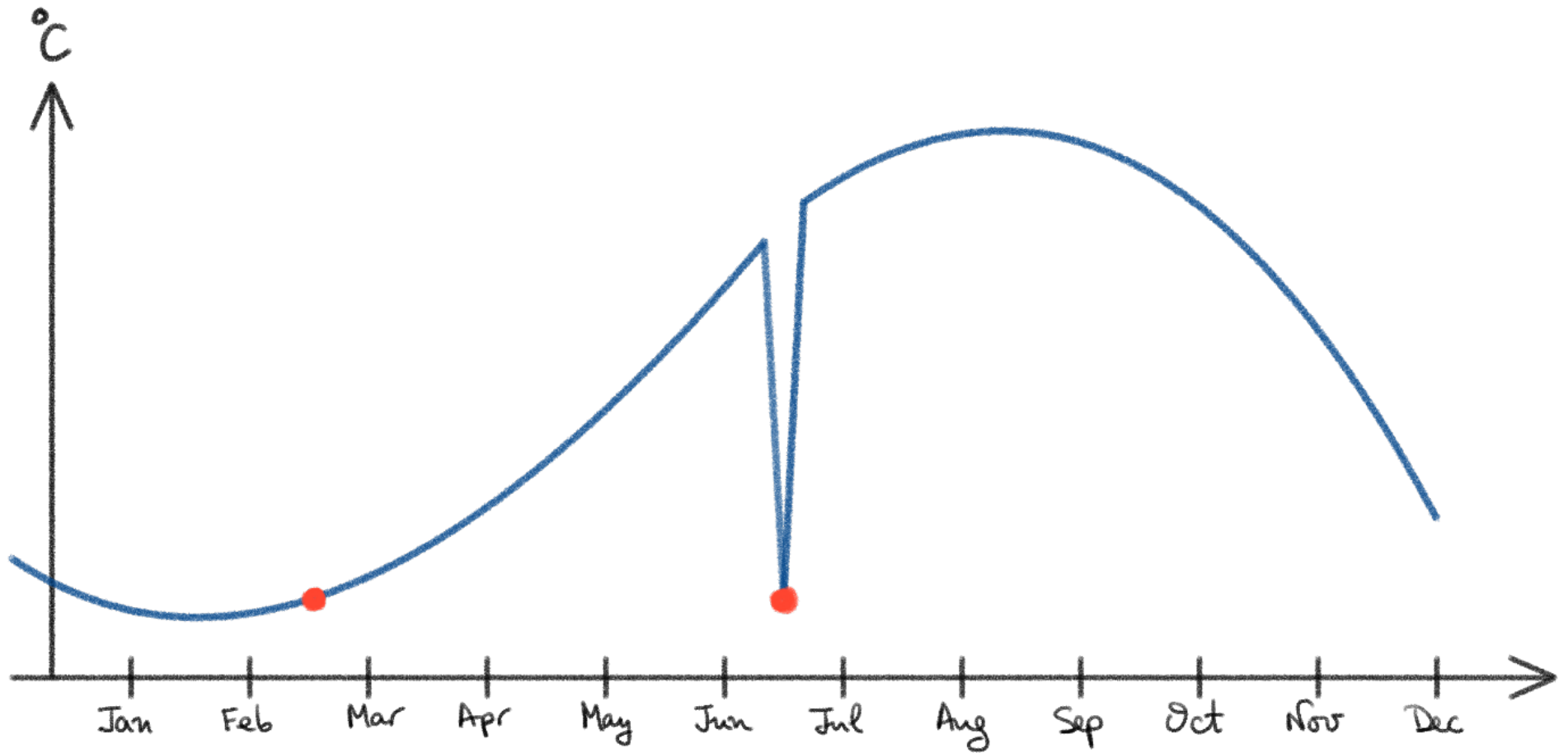
Traditionally:

→ Point anomalies

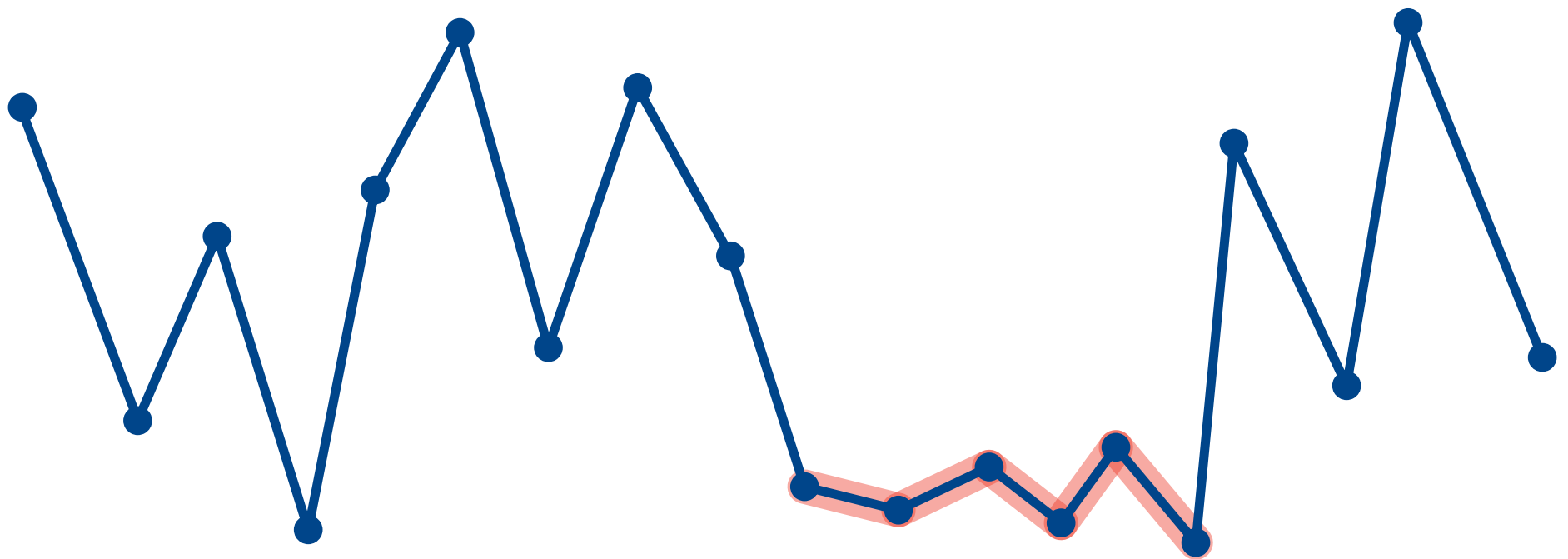
→ Conditional / Contextual anomalies

→ Group / Collective anomalies

CONTEXTUAL ANOMALIES



COLLECTIVE ANOMALIES



TYPES OF ANOMALIES

Since Deep Learning:

→ Low-level sensory anomalies

→ High-level semantic anomalies

LOW LEVEL, SENSORY



The book is
interesting and
it conveys a
meaningful message.

HIGH LEVEL, SEMANTIC



The girl was
playing with a
colorless green
liquid.

TYPES OF ANOMALIES

Anomaly not drawn from $\mathbb{P}^+$

Outlier drawn from $\mathbb{P}^+$ w low proba

Novelty from an evolving $\mathbb{P}^+$

ANOMALY DETECTION STRATEGIES

→ Model-agnostic

Given $X_1, \dots, X_n \sim \mathbb{P}^+$, $X_{\text{test}} \in \mathcal{A}_Z$?

→ Model-based

Given $(X_1, Y_1), \dots, (X_n, Y_n) \sim \mathbb{P}^+ \times \mathbb{P}_Y$

and $h: \mathcal{X} \rightarrow \mathcal{Y}$, $X_{\text{test}} \in \mathcal{A}_Z$?

ANOMALY DETECTION ALGORITHMS

AD algorithm

$s: X \rightarrow \mathbb{R}$ and $\tau \in \mathbb{R}$

such that

 s is the **score** and τ is the **threshold**

$\left\{ \begin{array}{l} \text{if } s(x) \leq \tau \Rightarrow x \text{ is declared "normal"} \\ \text{if } s(x) > \tau \Rightarrow x \text{ is declared "anomalous"} \end{array} \right.$

ANOMALY DETECTION ALGORITHMS

→ One-class classifier

→ Probability density estimation

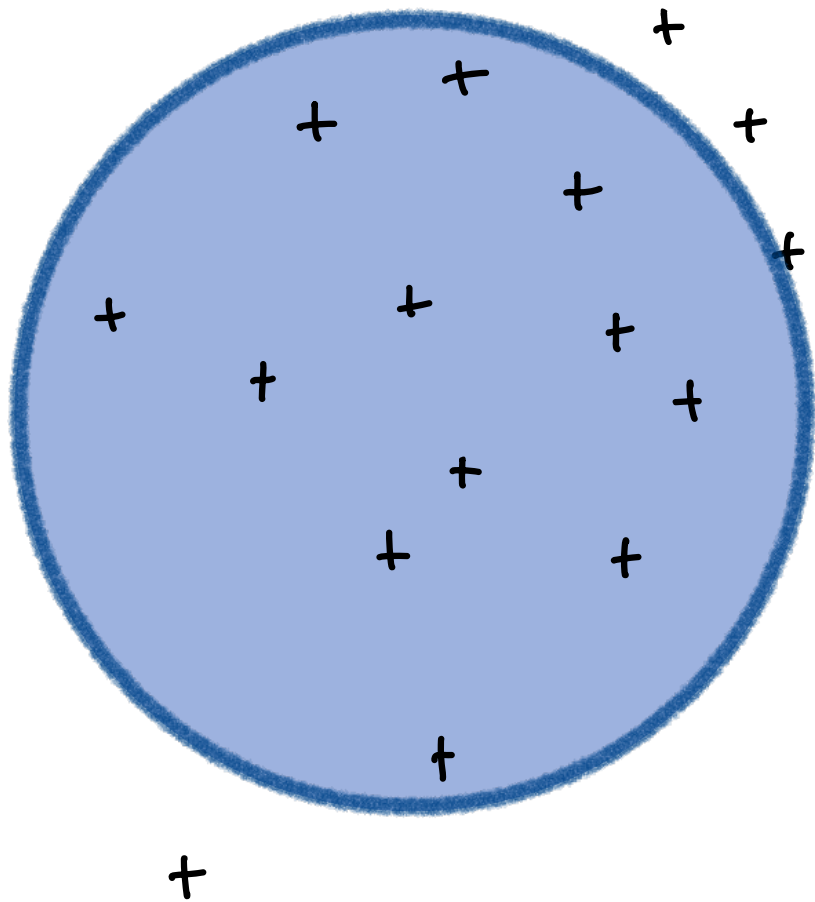
→ Reconstruction

→ Distance

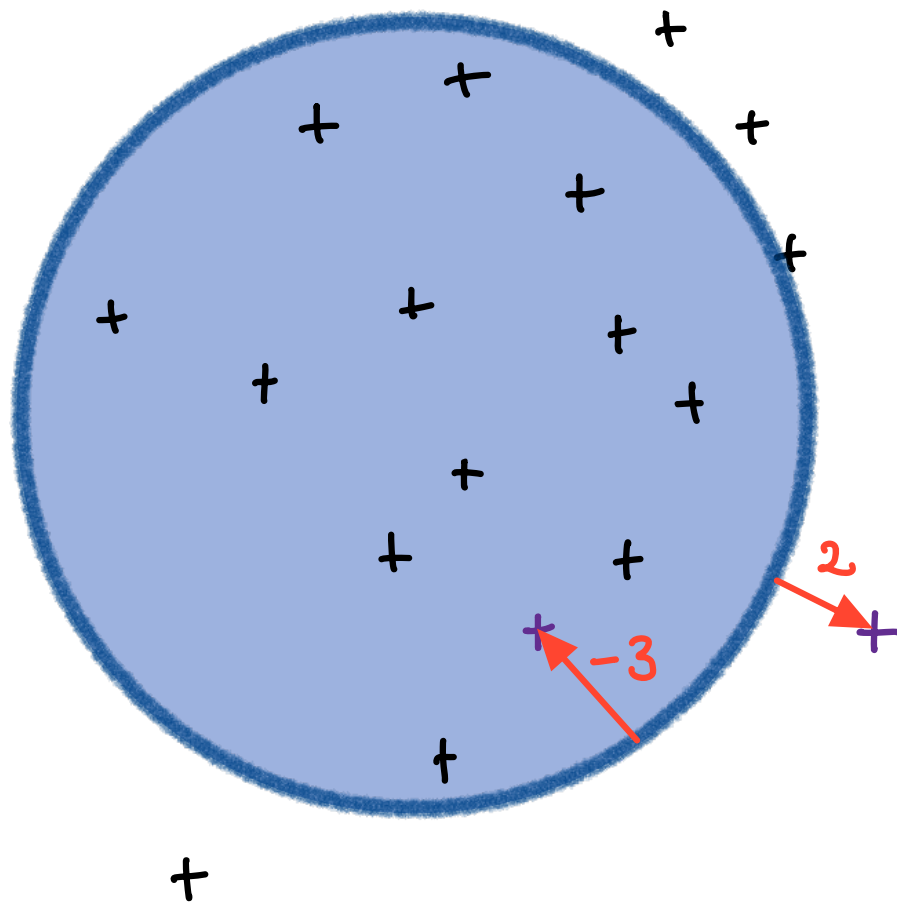
+ predictor $h: X \rightarrow Y$

(if available)

ONE-CLASS SVM



ONE-CLASS SVM



$S(x)$ = signed distance
from x to $\bigcirc$

$$z = 0$$

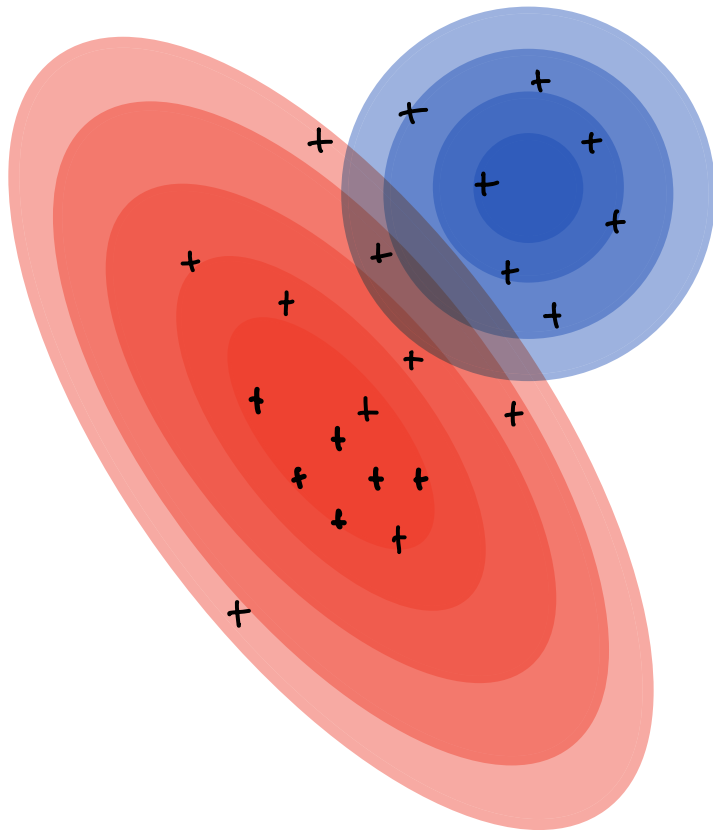
$$\Rightarrow \begin{cases} x \in \bigcirc \Rightarrow \text{"normal"} \\ x \notin \bigcirc \Rightarrow \text{"anomaly"} \end{cases}$$

PROBABILITY DENSITY ESTIMATION

e.g. with a Gaussian Mixture Model (GMM)

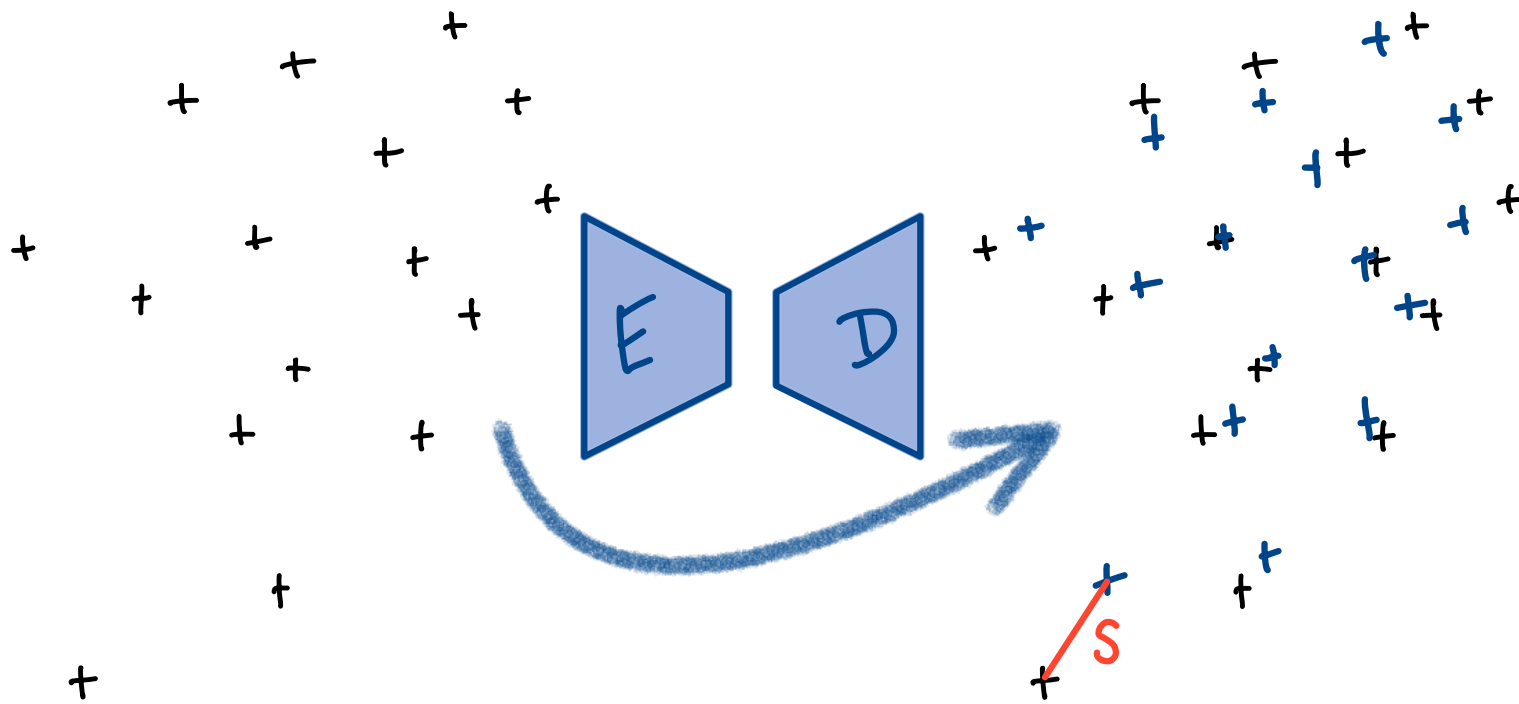
i.e. $\mathbb{P}^+ \simeq \sum_{i=1}^K w_i \mathcal{N}(\mu_i, \Sigma_i)$

GMM fitted on
 $x_1, \dots, x_n$



$S(x) = -\text{log-likelihood}$
of x under the GMM

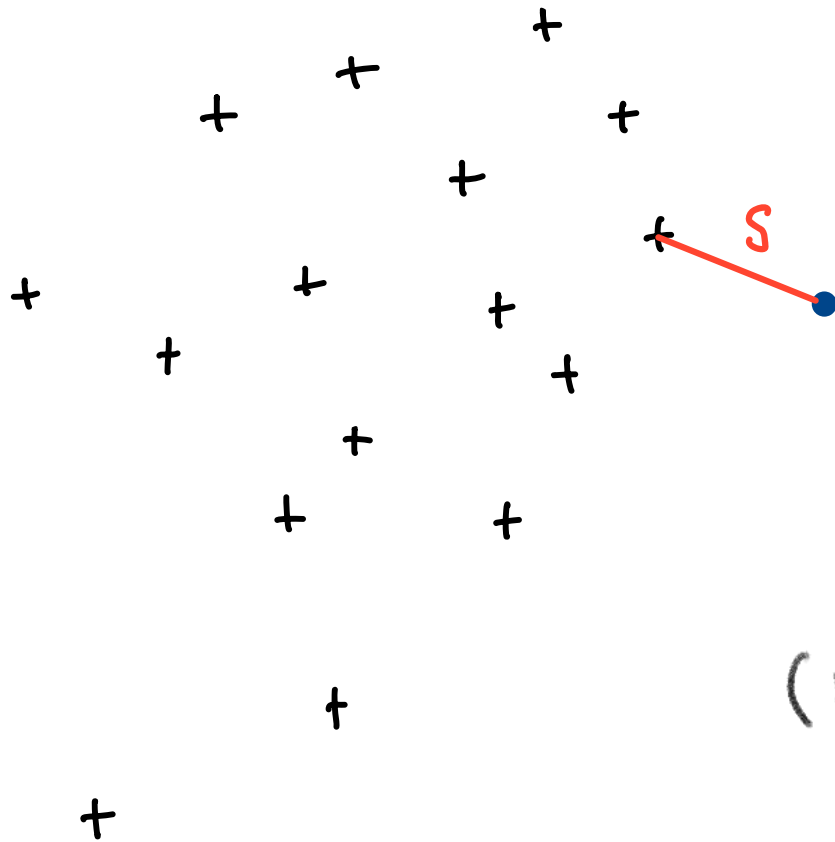
RECONSTRUCTION W. AUTOENCODERS



$$S(x) = d(x, \mathcal{D}(E(x)))$$

DISTANCE W. K-NEAREST NEIGHBORS

$$+ \sim \mathbb{P}^+$$



$S(x)$ = distance to
(K-th) nearest neighbor

Quiz

	sensitive to outliers?	scales to large dim?	easy to pick?
SVM			
GMM			
AE			
K-NN			

EVALUATION METRICS

Normal = -

Anomaly = +

		Predicted	
		-	+
Ground Truth	-	True Negative	False Positive
	+	False Negative	True Positive

Confusion Matrix

EVALUATION METRICS

For a fixed threshold z :

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

		Predicted	
		—	+
Ground Truth	—	True Negative	False Positive
	+	False Negative	True Positive

$$\text{Recall / True Positive Rate} = \frac{TP}{TP + FN}$$

EVALUATION METRICS

For a fixed threshold z :

$$F_1 \text{ score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$\text{False Positive Rate} = \frac{FP}{TN + FP}$$

Quiz

1. Is accuracy an adequate metric for AD? Why?
2. Give an example of a confusion matrix where precision and recall are both high.
3. Express the F_1 score in terms of the elements of the confusion matrix
4. The F_β score is
$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \beta^2 \text{Recall}}$$

What term do we give more importance to

(i) if $\beta < 1$? (ii) if $\beta > 1$?

EVALUATION METRICS

Threshold independent: AUC-ROC / AUROC

ROC = "Receiver Operating Characteristic"

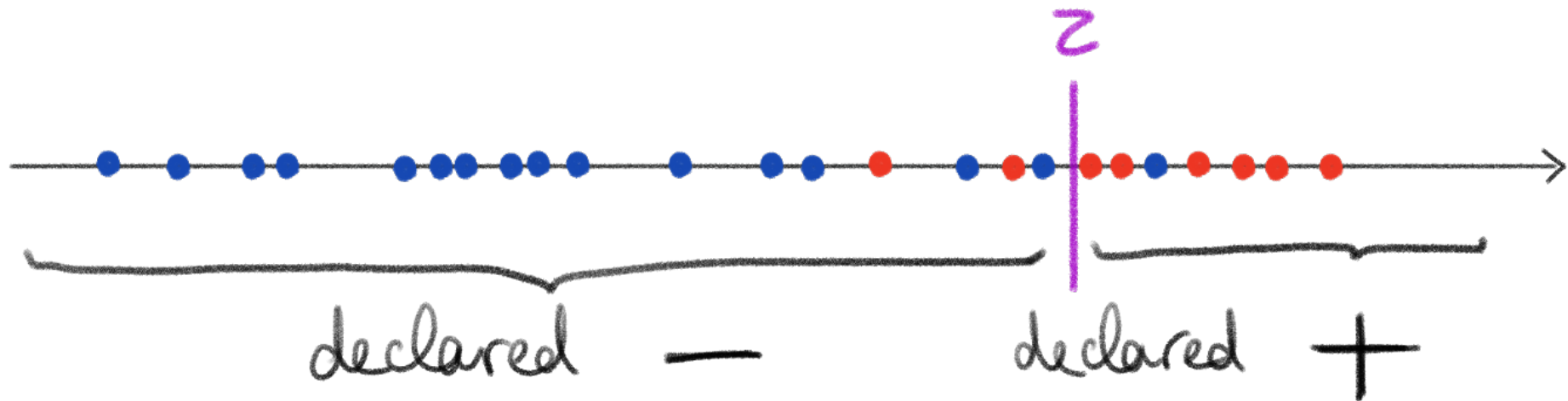
Given: $s: X \longrightarrow \mathbb{R}$

- = normal (ground-truth)

- = anomaly (ground-truth)



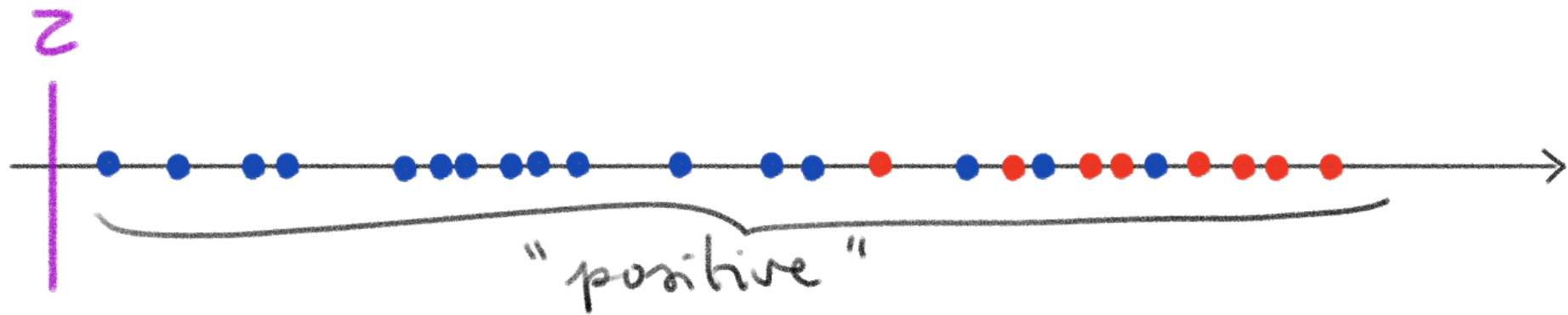
EVALUATION METRICS



$$TPR(z) = \frac{\# \text{ of } \bullet \text{ and } +}{\# \text{ of } \bullet}$$

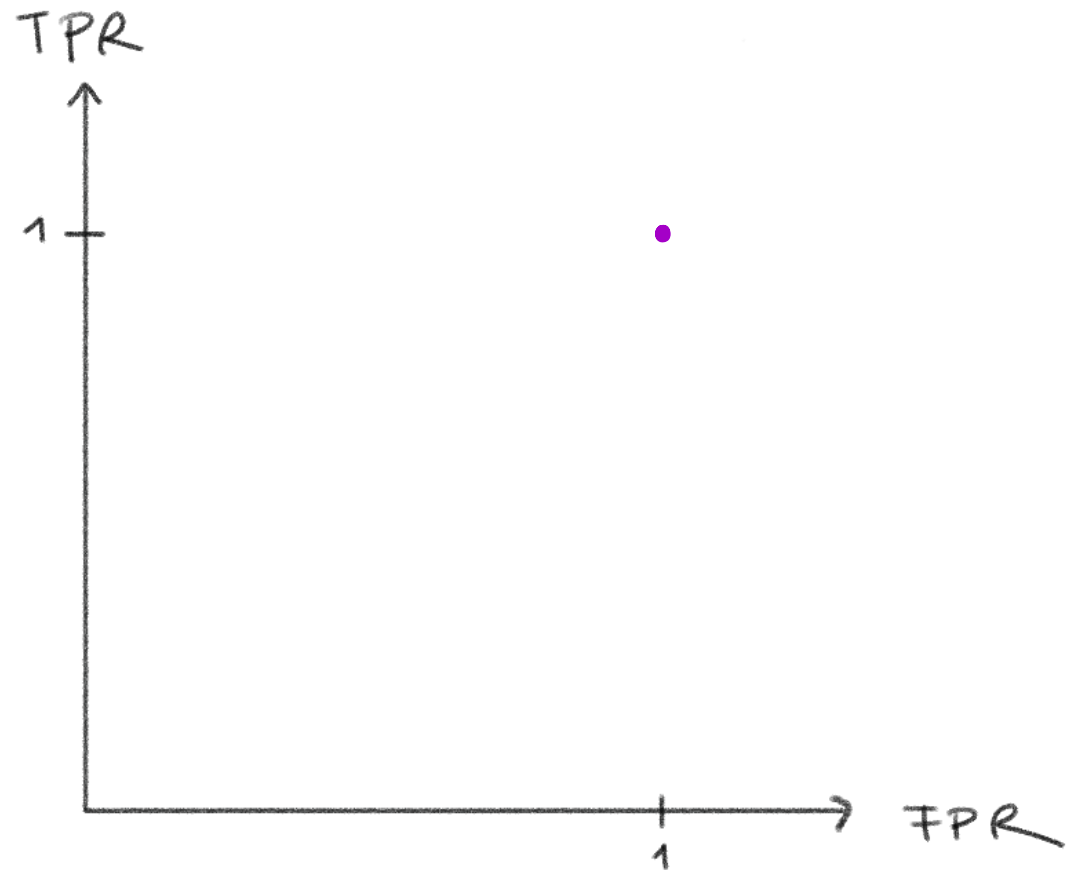
$$FPR(z) = \frac{\# \text{ of } \bullet \text{ and } -}{\# \text{ of } \bullet}$$

EVALUATION METRICS

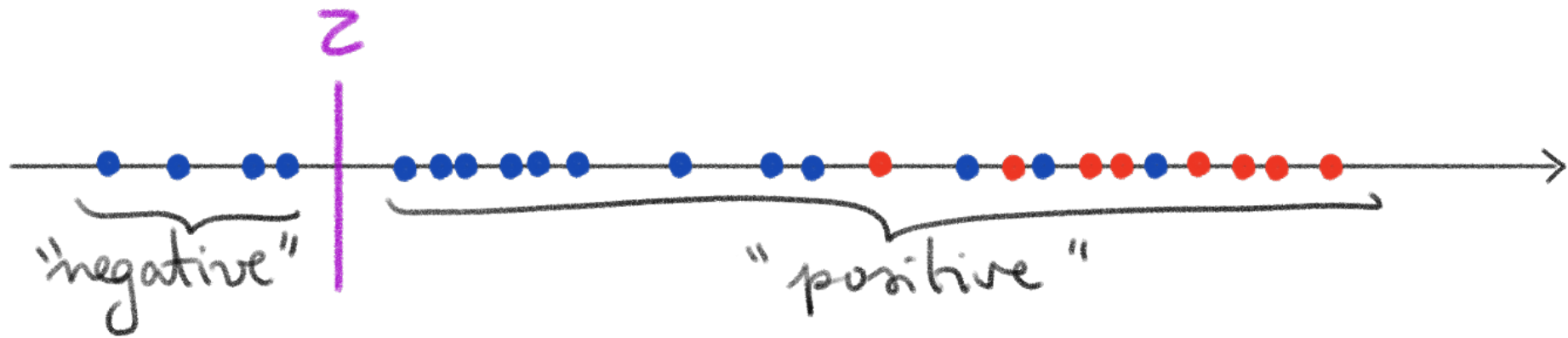


$$FPR(z) = 1$$

$$TPR(z) = 1$$

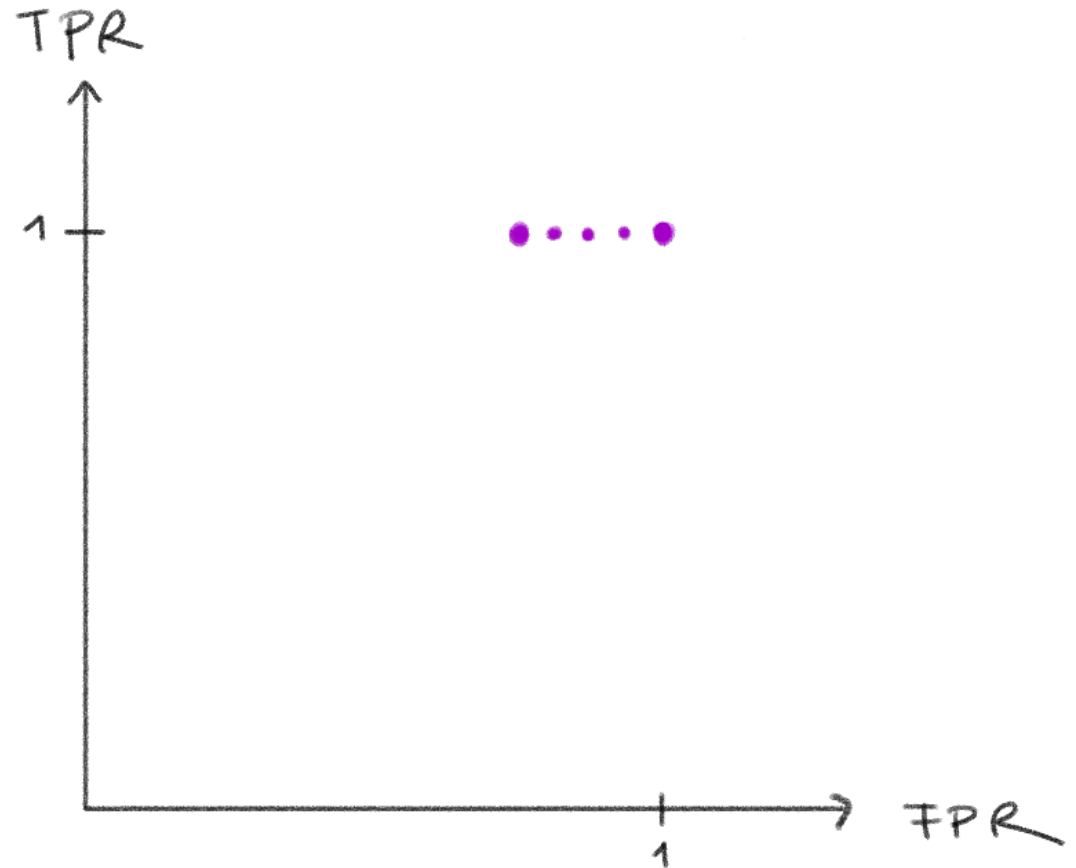


EVALUATION METRICS

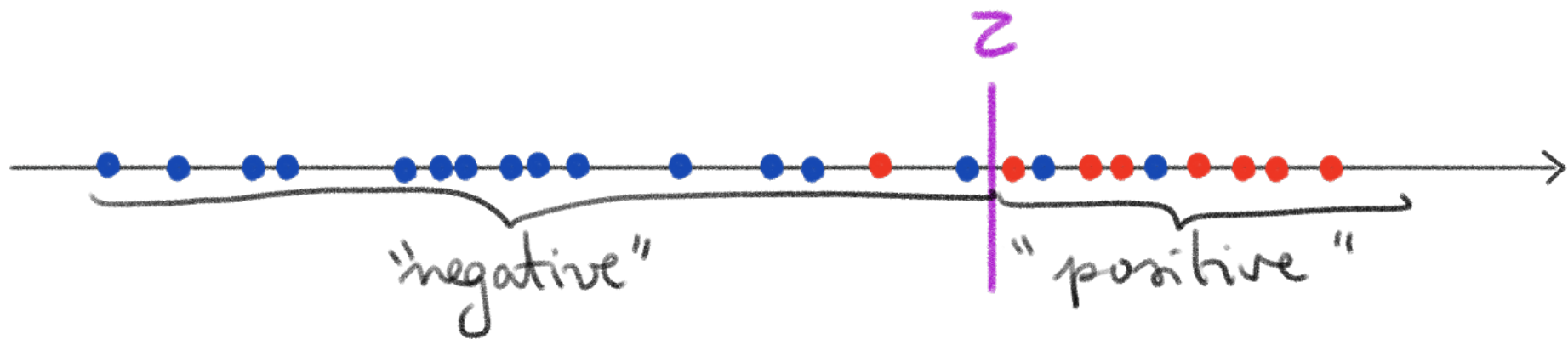


$$FPR(z) = \frac{12}{16} = \frac{3}{4}$$

$$TPR(z) = 1$$

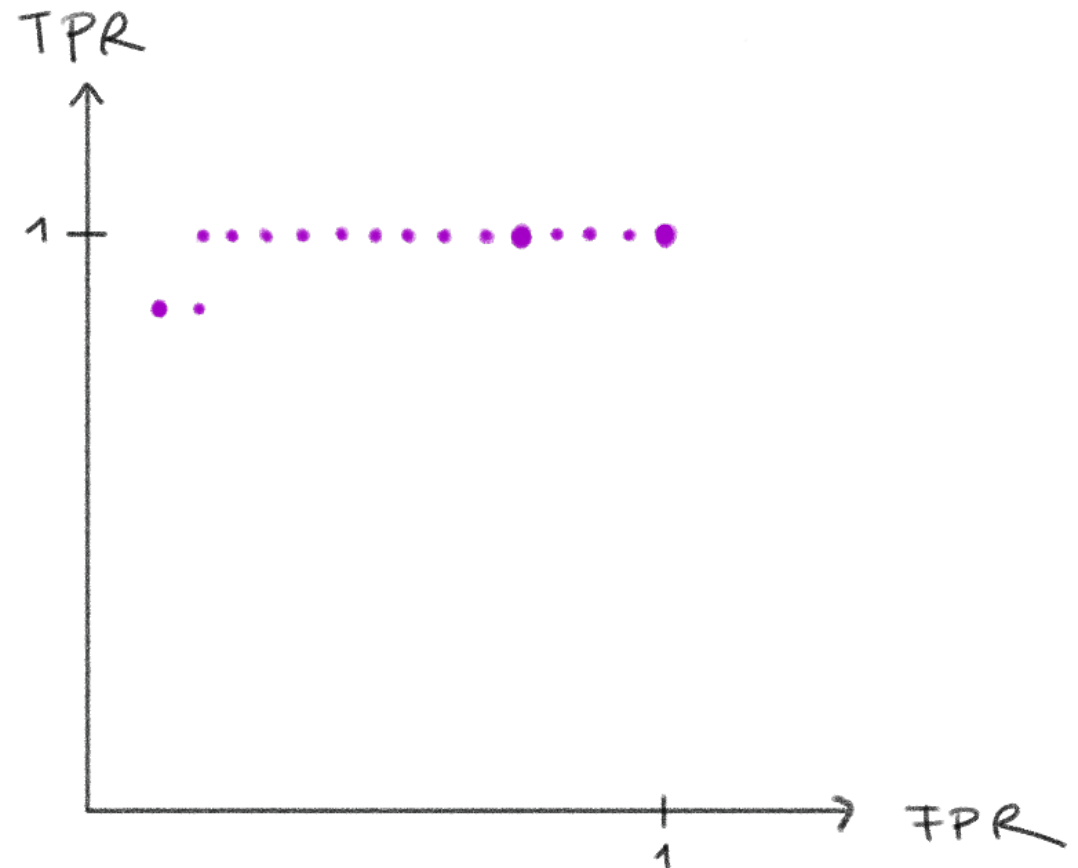


EVALUATION METRICS

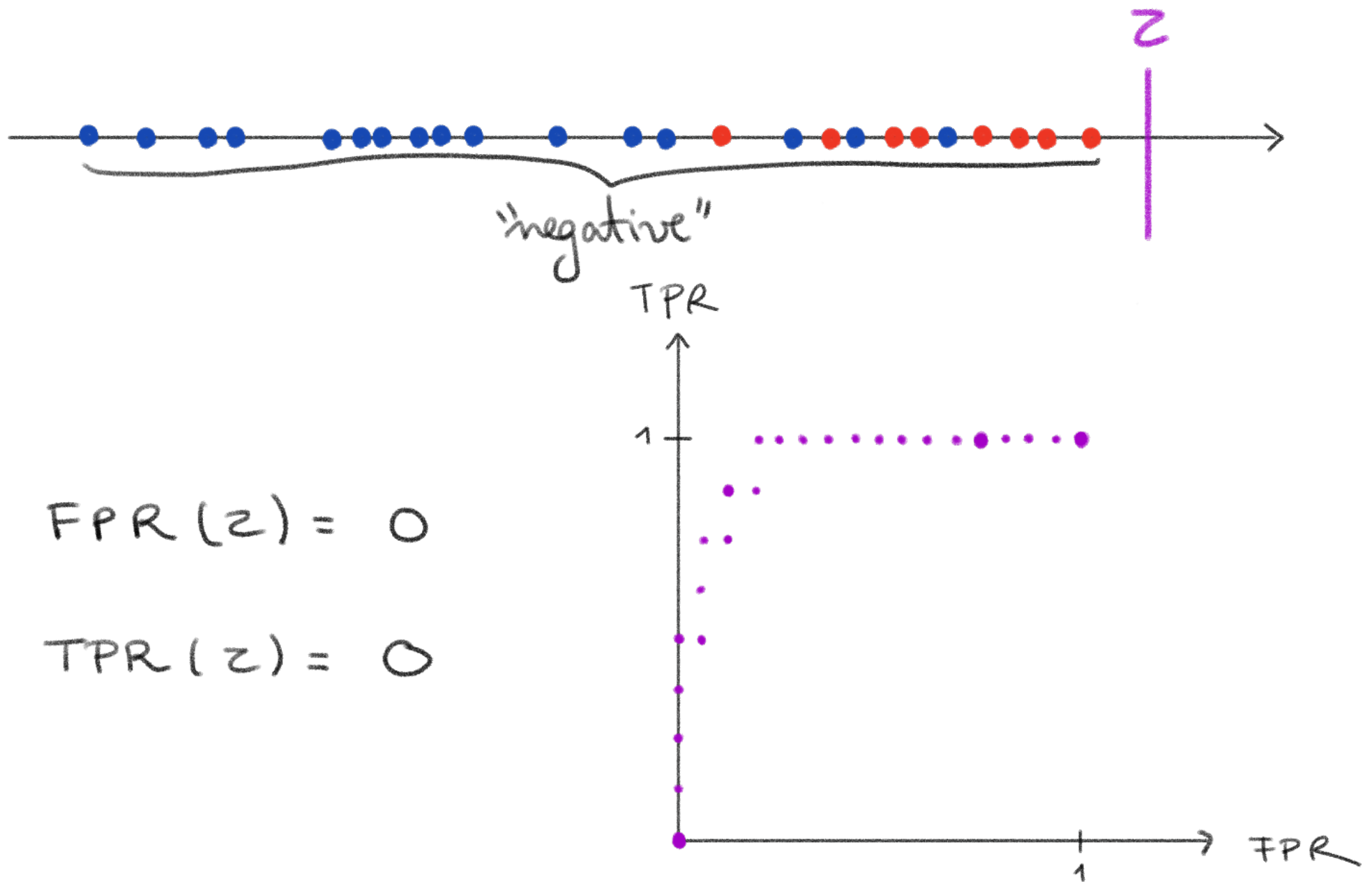


$$FPR(z) = \frac{1}{8}$$

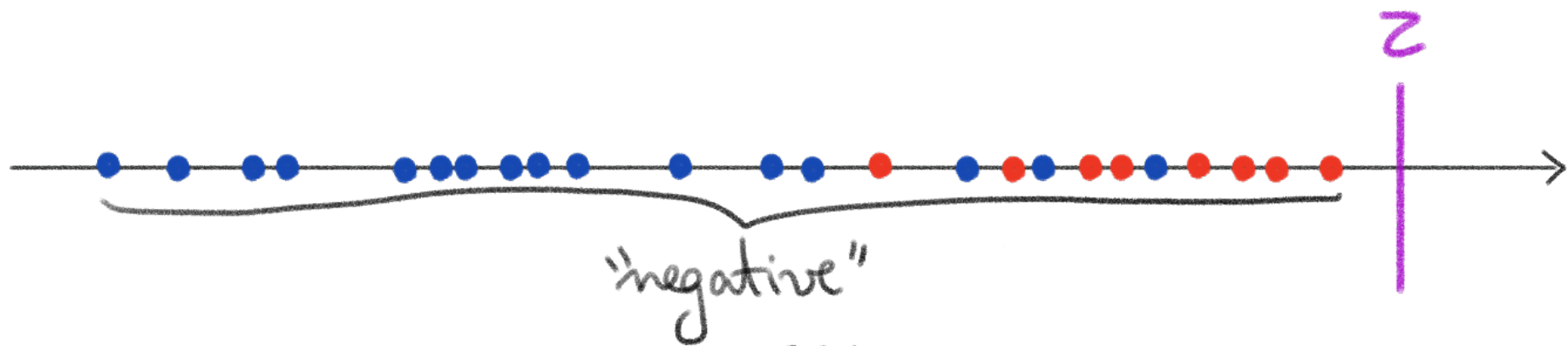
$$TPR(z) = \frac{7}{8}$$



EVALUATION METRICS

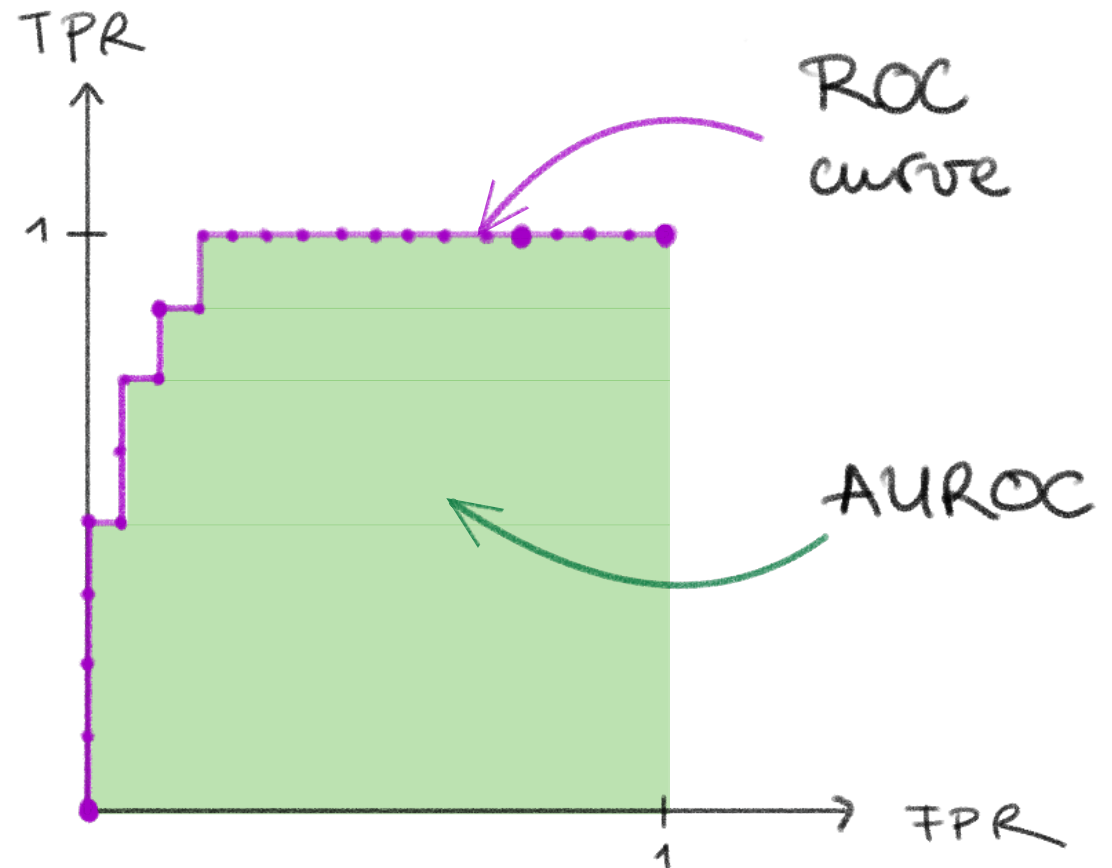


EVALUATION METRICS



$$FPR(z) = 0$$

$$TPR(z) = 0$$



Quiz

1. What would a "perfect" score look like?
And the ROC curve/AUROC associated?
2. Same question for a "random" score.
3. Same question for a "bad" score.
4. Can we generalize the ROC curve/AUROC to the case of probability distributions?

BENCHMARKING



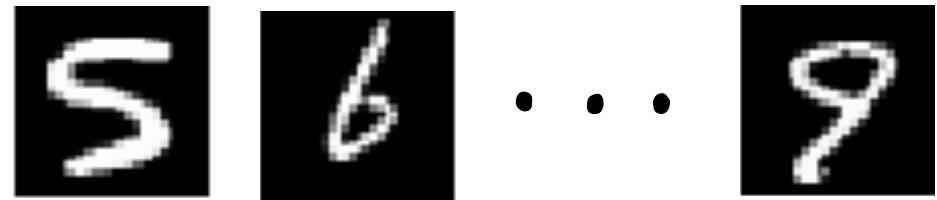
Lack of datasets for high-dimensional data (images, text)

see for example MVTecAD

Solution 1.



normal



anomalies

BENCHMARKING



Lack of datasets for high-dimensional data (images, text)

see for example MVTecAD

Solution 2 .

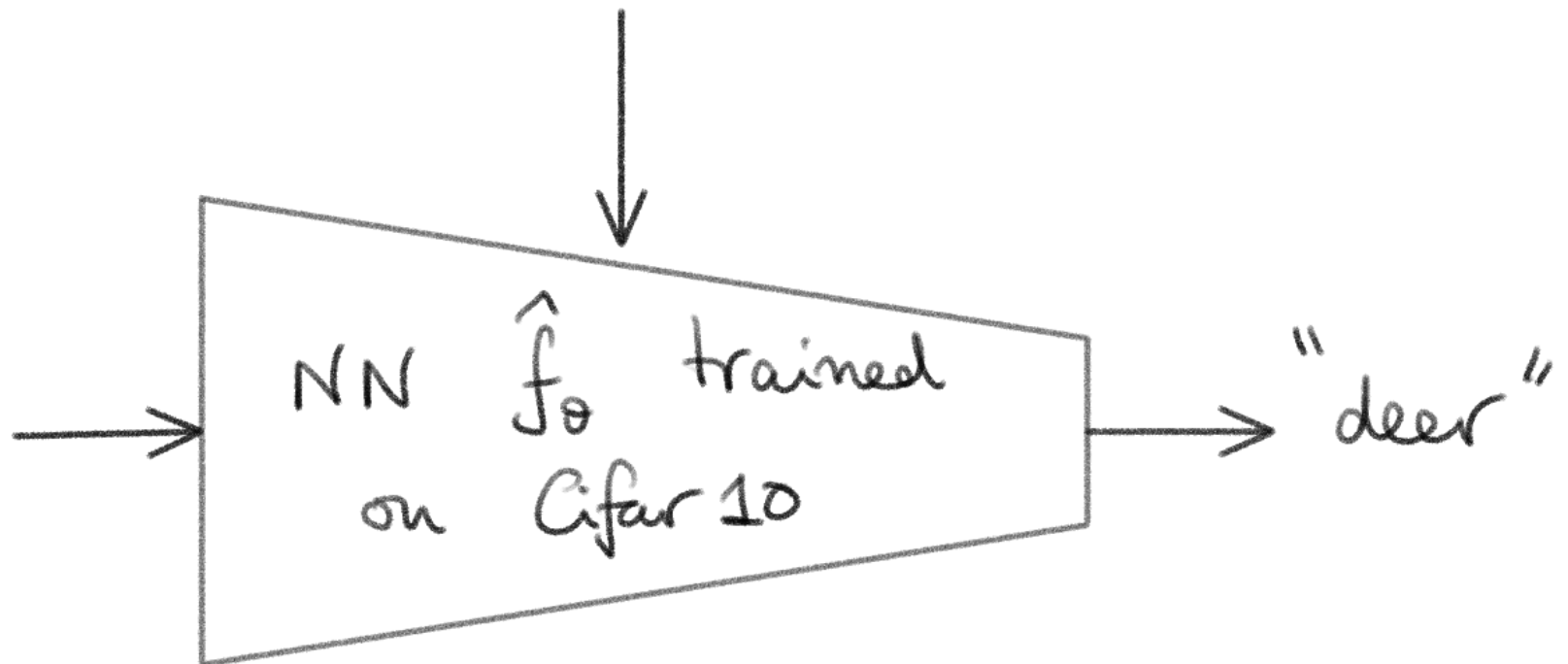
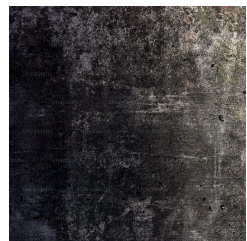


MNIST → normal



CIFAR10 → anomaly

OUT-OF-DISTRIBUTION DETECTION



OoD = AD ?

→ They accept similar mathematical formulations, but:

1. In OoD detection the space of anomalous images can be much larger
2. In OoD detection we have access to feature representations via $\hat{f}_\theta$

OoD SCORES

For a classifier $\hat{f}_\theta$:

$$\hat{f}_\theta : \mathcal{X} \xrightarrow{\hat{g}_\theta} \mathbb{R}^D \xrightarrow{\hat{h}_\theta} \mathbb{R}^d$$

feature space
(usually the penultimate layer)

logit space

→ logit-based scores: $s(x) = \tilde{S}(\hat{f}_\theta(x))$

→ feature-based scores: $s(x) = \tilde{S}(\hat{g}_\theta(x))$

LOGIT-BASED SCORES

→ Maximum Logit Score MLS

$$\tilde{S}(z) = -\max_{1 \leq c \leq C} z_c$$

→ Maximum Softmax Probability

$$\tilde{S}(z) = -\max_{1 \leq c \leq C} P_c(z)$$

LOGIT-BASED SCORES

→ Energy

$$\tilde{S}(z) =$$

$$T \log \sum_{c=1}^d e^{\beta_c / T}$$

tunable
hyperparameter

→ Entropy

$$\tilde{S}(z) = \sum_{c=1}^d P_c(z) \log P_c(z)$$

FEATURE-BASED SCORES

→ Deep K-nearest neighbor (DKNN)

Given $z_1, \dots, z_N$ and z_* :

↑
fitting dataset

↑
test point

(i) Compute $d_1 = d(z_1, z_*), \dots, d_N = d(z_N, z_*)$

(ii) Rank $d_{(1)} \leq d_{(2)} \leq \dots \leq d_{(N)}$

(iii) DKNN score $\tilde{S}(z_*) = d_{(k)}$

hyper-parameter
↑

FEATURE-BASED SCORES

→ Mahalanobis

Given $z_1, \dots, z_N$ and z_* :

↑
fitting dataset

↑
test point

(i) Fit Q Gaussians: $\mathcal{N}(\mu_c, \Sigma)$, $c=1, \dots, Q$

$$\hat{\mu}_c := \frac{1}{N_c} \sum_{i: y_i=c}^N z_i, \quad \hat{\Sigma} = \frac{1}{N} \sum_{c=1}^Q \sum_{i: y_i=c} (z_i - \hat{\mu}_c)(z_i - \hat{\mu}_c)^T$$

FEATURE-BASED SCORES

→ Mahalanobis

(ii) Mahalanobis score:

$$\tilde{S}(z) = \max_{1 \leq c \leq C} (z - \hat{\mu}_c)^T \hat{\Sigma}^{-1} (z - \hat{\mu}_c)$$

Exercise - If $C=1$, $\hat{\mu}_c = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\hat{\Sigma} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$,

draw $A \subseteq \mathbb{R}^2$ given by $A = \{z : \tilde{S}(z) \leq 1\}$

RESOURCES

Surveys

1. A unifying review of deep and shallow anomaly detection,
Ruff et al. 2021



2. Generalized out-of-distribution detection, Yang et al. 2021



RESOURCES

GitHub repositories

1. OpenOOD



2. ADBench



CHALLENGES

- Representation learning for AD
- Characterization of anomalies
- Interpretability of AD
- Creation of a unified methodology for AD research:
benchmarks, metrics,...

Kahoot
time!

Quiz

1. Why is AD important for ML/DL?
2. Name 2 classes of AD algorithms.
3. Write down the formula for 1 metric.
4. Write down 1 sentence stating 1 thing you understood well.
5. Write down 1 question with something you would like to know more about.